


# PROJECT DESIGN PRESENTATION

## FutureVerse: A Machine Learning-Based Framework for Sugarcane Growth Prediction and Intelligent Farming Recommendation

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| **GROUP NO:** | **A18** |

GUIDE : Ms. V. V. Manglaram

# FutureVerse

**A Machine Learning-Based Framework for Sugarcane Growth Prediction and Intelligent Farming Recommendations**

## Problem Statement

To design and develop an intelligent farming system that:
-Predicts sugarcane growth (%) and estimated yield.
-Uses soil, weather, irrigation, and crop parameters.
-Provides data-driven fertilizer and irrigation recommendations.
-Supports better crop and soil management decisions.

## Proposed Solution

To develop a machine learning-based decision-support system that:
-Applies multiple ML regression models for prediction
-Analyzes relevant agricultural parameters
-Selects the best-performing model using evaluation metrics
-Provides intelligent and practical farming recommendations

## Team Members

- Abhishek Danole — A-21
- Harsh Abad — A-01
- Paras Danole — A-22

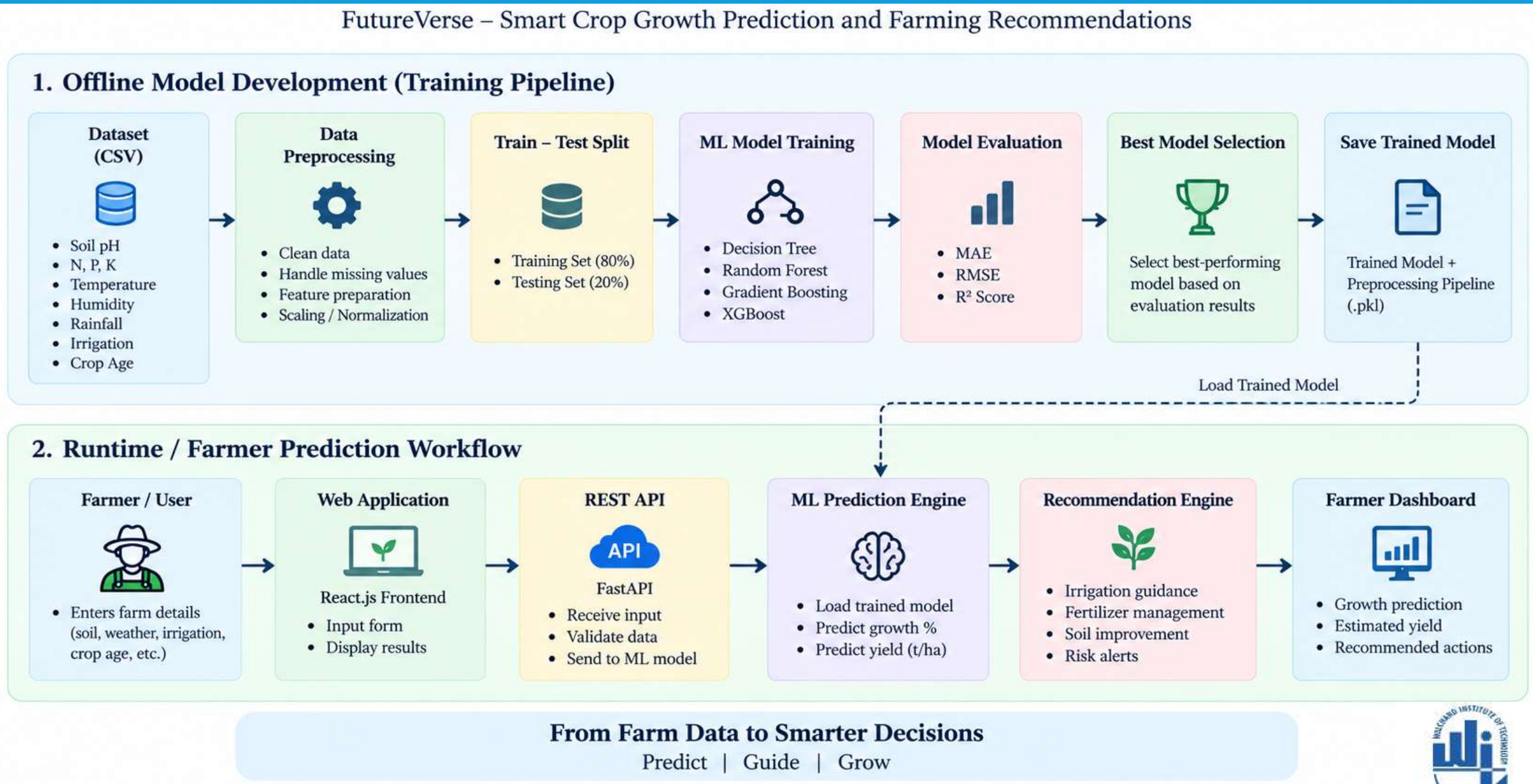
## Project Guide

- Ms. V.V. Manglaram
- Assistant Professor

## Expected Outcome

- Farmer-friendly web decision-support system
- Prediction + practical recommendations

# 2. OVERALL SYSTEM ARCHITECTURE



# 3. MODULE DESIGN

## 1 User Input Module

Collect soil pH, N, P, K, temperature, humidity, rainfall, irrigation and crop age.

## 2 Validation & Preprocessing

Validate values, handle missing data, normalize numerical features and select relevant inputs.

## 3 ML Prediction Engine

Train/use Decision Tree, Random Forest, Gradient Boosting and XGBoost regression models.

## 4 Best Model Selection

Evaluate MAE, RMSE and $R^2$; select the best-performing model on validation data.

## 5 Recommendation Engine

Generate fertilizer, irrigation, crop-management and soil-improvement guidance.

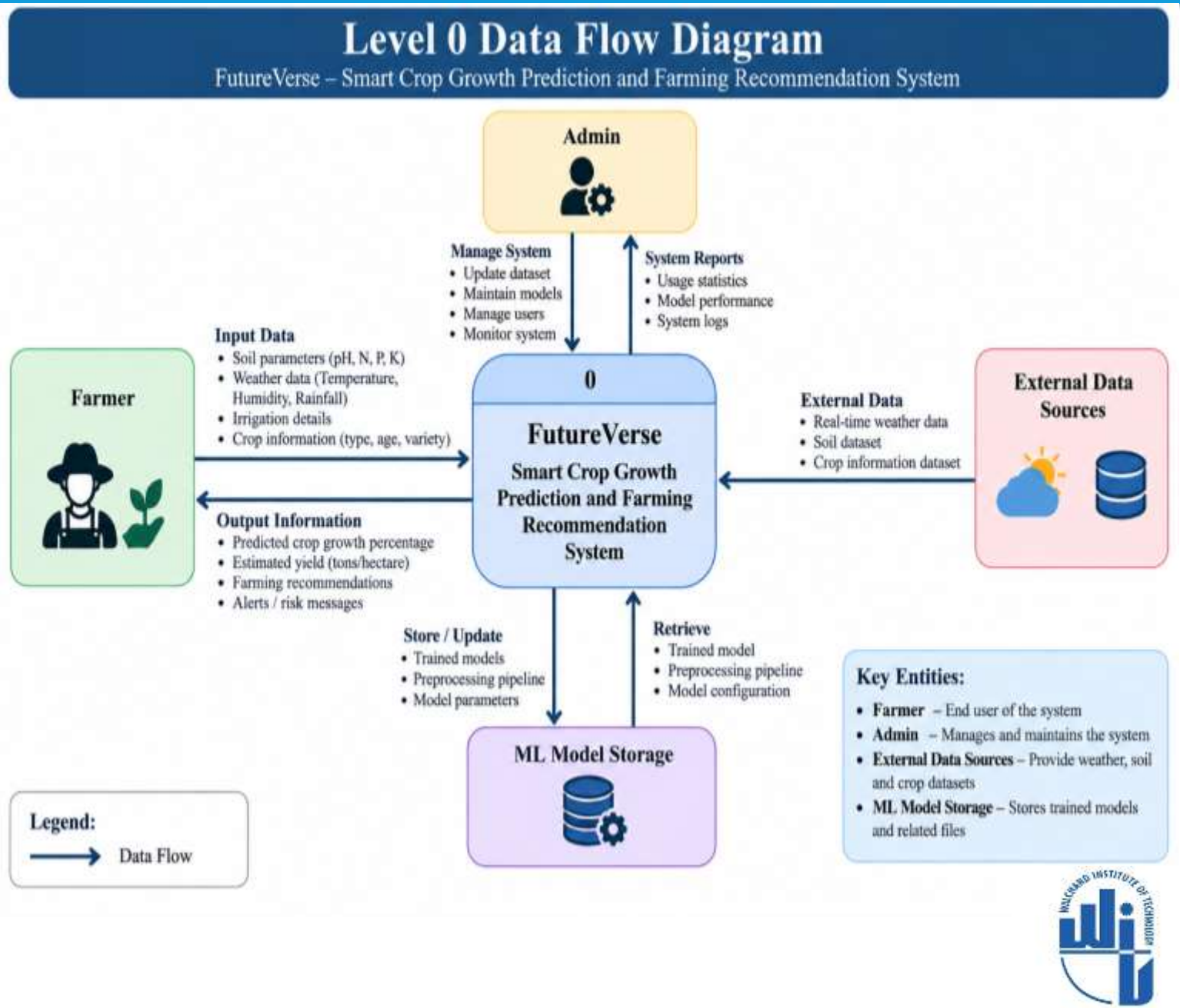
## 6 Result Generation

Display predicted growth, estimated yield and recommendations through the dashboard.

# 4. DATA / DATABASE / WORKFLOW DESIGN

## DATA-PROCESSING PIPELINE

| | | |
|---|---|---|
| 1 | **Load Dataset** | Historical agricultural records |
| 2 | **Clean & Validate** | Check ranges and missing values |
| 3 | **Prepare Features** | Normalize / select relevant inputs |
| 4 | **Split Data** | Training and validation/test sets |
| 5 | **Train Models** | Four regression algorithms |
| 6 | **Evaluate & Select** | MAE, RMSE, $R^2$ |
| 7 | **Save Pipeline** | Selected model + preprocessing |
| 8 | **Predict & Recommend** | Growth, yield and actions |



# 5. CORE TECHNICAL DESIGN

## Machine Learning Prediction

1. Prepare input features from soil, weather, irrigation and crop age.
2. Train four regression models:
• Decision Tree Regressor
• Random Forest Regressor
• Gradient Boosting Regressor
• XGBoost Regressor
3. Evaluate using MAE, RMSE and $R^2$.
4. Select the best-performing model empirically.
5. Predict Growth (%) and Estimated Yield (t/ha).

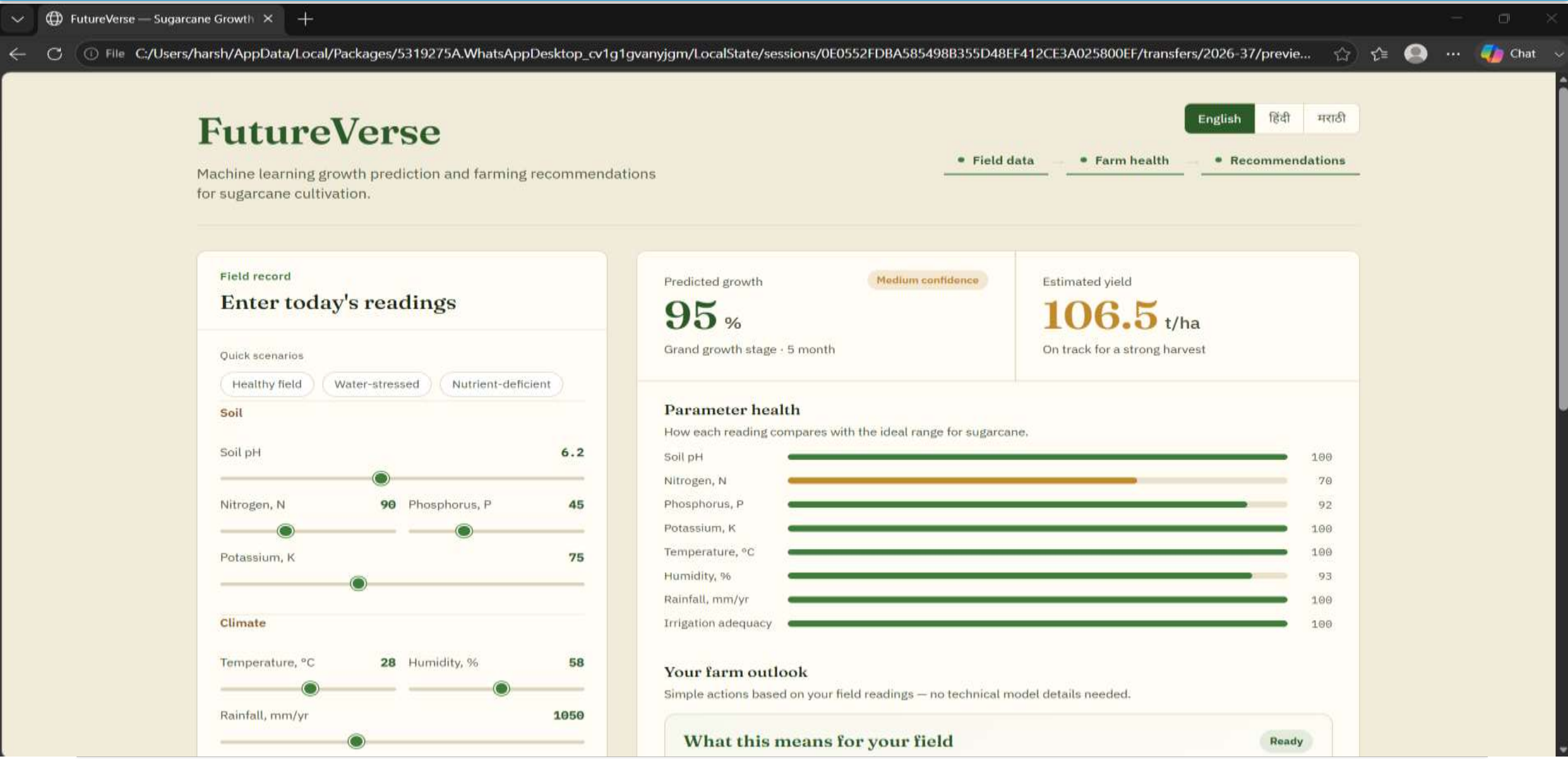
## Intelligent Recommendation Engine

Prediction results + field conditions
• → Fertilizer application guidance
• → Irrigation scheduling guidance
• → Crop management practices
• → Soil improvement strategies

## REST API & Runtime Flow

• React frontend sends farmer inputs to the REST API.
• Backend validates and preprocesses the request using the trained pipeline.
• Selected ML model generates prediction.
• Recommendation engine converts results into understandable actions.
• Dashboard displays the final output.

# 6. UI / PROTOTYPE / WIREFRAMES



FutureVerse — Sugarcane Growth
File C:/Users/harsh/AppData/Local/Packages/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/0E0552FDBA585498B355D48EF412CE3A025800EF/transfers/2026-37/previe...
Chat

Rainfall, mm/yr **1050**

**Irrigation & crop**

Irrigation adequacy

Adequate

Soil type

Clay

Crop age, months **5**

**Analyze growth**

Simple actions based on your field readings — no technical model details needed.

## What this means for your field

Ready

Focus on the areas that can improve your crop first.

| FIELD STATUS | TOP PRIORITY | NEXT STEP |
|---|---|---|
| – | – | – |

## Recommendations

Generated from the gap between current readings and target ranges for sugarcane.

### Fertilizer

Apply a balanced NPK dose to raise nitrogen — currently 90 kg/ha, target 120–180; and add phosphorus — currently 45 kg/ha, target 50–80.

### Irrigation

clay soil retains water — space out irrigation to avoid waterlogging.

### Crop management

the crop is in its grand growth stage; this is the main period for nutrient uptake — keep fertilizer and water on schedule.

### Soil improvement

soil pH (6.2) is within the ideal range for sugarcane. Adding organic compost or farmyard manure will improve long-term soil structure and nutrient retention.

Download report

Readings are processed for this session only and are not stored. Figures are illustrative estimates for demonstration purposes, not agronomic guarantees.

# 7. TECHNOLOGY STACK & IMPLEMENTATION STATUS

## Frontend

- React.js
- JavaScript ES6+
- HTML5
- CSS3
- Bootstrap
- Interactive dashboard

## Backend & ML

- Python 3.11
- FastAPI / RESTful API
- Scikit-learn
- XGBoost
- Pandas & NumPy
- Matplotlib

## Data / Platform

- CSV / dataset files for training
- Runtime processing of farmer inputs
- No traditional database required for core workflow
- Windows 11 / Ubuntu 22.04
- Client-server architecture

## Implementation Status

- Farmer-friendly UI prototype prepared
- Field Data, Farm Outlook & Recommendation screens designed
- Data set is finalized for train model
- ML/API integration in progress

## Immediate Next Milestone

- Train ML models
- Connect trained model with FastAPI
- Integrate predictions with recommendation engine
- End-to-end testing and deployment

# QUESTIONS

# THANK YOU !